

N-Methyl-2-Pyrrolidone (NMP)

PRD / SDSSG / EN / 0001

Version	Revision Date:	SDS Number:	Date of last issue: 05.12.2022
1.5	23.01.2025	150000103627	Date of first issue: 22.10.2015

Section 1: Identification

Product identifier : N-Methyl-2-Pyrrolidone (NMP)

Product code : 51059-00, C5105902, C5105901, E5105901, N5105911, N5105912, N5105913, N5105914, P5105930, P5105921, P5105920, P5105918, P5105917, P5105916, P5105931, C5105903, P5105934, P5105939, P5105938, P5105933, P5105935, P5105932

Recommended use of the chemical and restrictions on use

Recommended use : Solvent

Restrictions on use : None known.

Manufacturer or supplier's details

Company : Eastman Chemical Company

Address : 200 South Wilcox Drive
Kingsport TN 37660-5147

Telephone : (423) 229-2000

Emergency telephone number : NCEC +65 6262-6462

Section 2: Hazard identification**Classification of the substance or mixture**

Skin corrosion/irritation : Category 2

Serious eye damage/eye irritation : Category 2A

Reproductive toxicity : Category 1B

Specific target organ toxicity - single exposure : Category 3 (Respiratory system)

GHS Label elements, including precautionary statements

Hazard pictograms :  

Signal word : Danger

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Hazard statements : H315 Causes skin irritation.
H319 Causes serious eye irritation.
H335 May cause respiratory irritation.
H360 May damage fertility or the unborn child.

Precautionary statements : **Prevention:**
P201 Obtain special instructions before use.
P202 Do not handle until all safety precautions have been read and understood.
P261 Avoid breathing mist or vapours.
P264 Wash skin thoroughly after handling.
P271 Use only outdoors or in a well-ventilated area.
P280 Wear protective gloves/ protective clothing/ eye protection/ face protection.

Response:

P302 + P352 IF ON SKIN: Wash with plenty of water.
P304 + P340 + P312 IF INHALED: Remove person to fresh air and keep comfortable for breathing. Call a POISON CENTER/ doctor if you feel unwell.
P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P308 + P313 IF exposed or concerned: Get medical advice/ attention.
P332 + P313 If skin irritation occurs: Get medical advice/ attention.
P337 + P313 If eye irritation persists: Get medical advice/ attention.
P362 + P364 Take off contaminated clothing and wash it before reuse.

Storage:

P403 + P233 Store in a well-ventilated place. Keep container tightly closed.
P405 Store locked up.

Disposal:

P501 Dispose of contents/ container to an approved waste disposal plant.

Other hazards which do not result in classification

None known.

Section 3: Composition/information on ingredients

Substance / Mixture : Substance

Components

Chemical name	CAS-No.	Concentration (% w/w)
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N-methyl pyrrolidone	872-50-4	>= 90 -<= 100
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Eastman is committed to the safety, health and environment of our employees, our customers, and the communities we operate within. As part of this commitment, Eastman's Safety Data Sheets (SDS) are prepared in accordance with all applicable national and local regulations. The compositions of our documents reflect these requirements which include, but are not limited to, requirements under the Globally Harmonized System of Classification and Labeling (GHS). These compositions commonly involve the use of ranges versus specific analytical values. If you require a composition that is more specific, please refer to the Certificate of Analysis, sales specification, or contact your Customer Service Representative.

Section 4: First-aid measures**Description of necessary first-aid measures**

If inhaled	: Move to fresh air. Call a physician or poison control centre immediately.
In case of skin contact	: Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Wash contaminated clothing before re-use. Get medical attention. Thoroughly clean shoes before reuse.
In case of eye contact	: Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical advice/ attention.
If swallowed	: Seek medical advice.

Most important symptoms and effects, both acute and delayed

Risks	: None known.
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Indication of any immediate medical attention and special treatment needed

Treatment	: Treat symptomatically.
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Section 5: Fire-fighting measures**Extinguishing media**

Suitable extinguishing media	: Alcohol-resistant foam Carbon dioxide (CO2) Dry chemical Water spray
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Unsuitable extinguishing media	: Water spray jet
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Special hazards arising from the substance or mixture

Specific hazards during fire-fighting : Water may be ineffective.
The product will float on water and can be reignited on surface water.

Hazardous combustion products : No hazardous combustion products are known

Special protective actions for fire-fighters

Special protective equipment for firefighters : Wear an approved positive pressure self-contained breathing apparatus in addition to standard fire fighting gear.

Specific extinguishing methods : Use water spray to cool unopened containers.

Section 6: Accidental release measures**Personal precautions, protective equipment and emergency procedures**

Personal precautions : Wear appropriate personal protective equipment.
Local authorities should be advised if significant spillages cannot be contained.

Environmental precautions

Environmental precautions : Avoid release to the environment.

Methods and materials for containment and cleaning up

Methods for cleaning up : Contain spillage, soak up with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and transfer to a container for disposal according to local / national regulations (see section 13).
After cleaning, flush away traces with water.
Eliminate all ignition sources if safe to do so.

Section 7: Handling and storage**Precautions for safe handling**

Advice on protection against fire and explosion : None known.

Advice on safe handling : Avoid inhalation of vapor or mist.
Do not get on skin or clothing.
Do not get in eyes.
Avoid contact with skin, eyes and clothing.
Do not swallow.
Ensure adequate ventilation.
Wash thoroughly after handling.
Keep away from fire (No Smoking).

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Keep away from fire, sparks and heated surfaces.
Do not use sparking tools.

Hygiene measures : Handle in accordance with good industrial hygiene and safety practice.

Conditions for safe storage, including any incompatibilities

Conditions for safe storage : Keep container closed when not in use.
Store locked up.

Section 8: Exposure controls/personal protection**Control parameters****Occupational Exposure Limits**

Contains no substances with occupational exposure limit values.

Appropriate engineering control measures : Ensure adequate ventilation.

Individual protection measures, such as personal protective equipment (PPE)

Eye/face protection : Wear safety glasses with side shields (or goggles).
Face-shield
Always wear eye protection when the potential for inadvertent eye contact with the product cannot be excluded.

Respiratory protection : Use respiratory protection unless adequate local exhaust ventilation is provided or exposure assessment demonstrates that exposures are within recommended exposure guidelines.

Hand protection

Remarks : Wear suitable gloves.

Protective measures : Remove respiratory and skin/eye protection only after vapors have been cleared from the area.
Ensure that eye flushing systems and safety showers are located close to the working place.
Use personal protective equipment as required.

Section 9: Physical and chemical properties

Appearance : liquid

Colour : colourless

Odour : amine-like

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Odour Threshold	:	not determined
pH	:	8.5 - 10 (20 °C) Concentration: 100 g/l
Melting point/freezing point	:	-24.2 °C
Boiling point/boiling range	:	204 °C
Flash point	:	91 °C Method: closed cup
Evaporation rate	:	not determined
Upper explosion limit / Upper flammability limit	:	9.5 %(V)
Lower explosion limit / Lower flammability limit	:	1.3 %(V)
Vapour pressure	:	0.32 hPa (20 °C)
Relative vapour density	:	3.42 (Air = 1.0)
Relative density	:	No data available
Density	:	1.03 g/cm ³ (25 °C)
Solubility(ies) Water solubility	:	completely soluble
Partition coefficient: n-octanol/water	:	log Pow: -0.46 (25 °C)
Auto-ignition temperature	:	245 °C
Decomposition temperature	:	not determined
Viscosity Viscosity, dynamic	:	1.661 mPa.s (25 °C)
Viscosity, kinematic	:	not determined
Explosive properties	:	Not explosive
Oxidizing properties	:	The substance or mixture is not classified as oxidizing.

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Molecular weight : 99.13 g/mol

Section 10: Stability and reactivity

Reactivity	: Stable
Chemical stability	: Stable under normal conditions.
Possibility of hazardous reactions	: Stable Hazardous decomposition products formed under fire conditions.
Conditions to avoid	: Heat, flames and sparks.
Incompatible materials	: Oxidizing agents
Hazardous decomposition products	: Carbon dioxide (CO ₂) Carbon monoxide

Section 11: Toxicological information**Acute toxicity**

Not classified based on available information.

Product:

Acute oral toxicity	: Remarks: No data available
Acute inhalation toxicity	: Remarks: No data available
Acute dermal toxicity	: Remarks: No data available

Components:**N-methyl pyrrolidone:**

Acute oral toxicity	: LD50 Oral (Rat): 4,150 mg/kg
Acute inhalation toxicity	: LC50 (Rat): > 5.1 mg/l Exposure time: 4 h Assessment: The substance or mixture has no acute inhalation toxicity
Acute dermal toxicity	: LD50 Dermal (Rat): > 5,000 mg/kg Assessment: The substance or mixture has no acute dermal toxicity

Skin corrosion/irritation

Causes skin irritation.

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Product:

Remarks : No data available

Components:**N-methyl pyrrolidone:**

Result : Irritating to skin.

Serious eye damage/eye irritation

Causes serious eye irritation.

Product:

Remarks : No data available

Components:**N-methyl pyrrolidone:**

Species	:	Rabbit
Result	:	Irritating to eyes.
Assessment	:	Causes serious eye irritation.

Respiratory or skin sensitisation**Skin sensitisation**

Not classified based on available information.

Respiratory sensitisation

Not classified based on available information.

Product:

Remarks : No data available

Components:**N-methyl pyrrolidone:**

Result : Did not cause sensitization on laboratory animals.

Germ cell mutagenicity

Not classified based on available information.

Components:**N-methyl pyrrolidone:**

Genotoxicity in vitro	:	Test Type: Salmonella typhimurium assay (Ames test) Result: negative
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Genotoxicity in vivo	:	Test Type: Mammalian Erythrocyte Micronucleus Test Result: negative
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Carcinogenicity

Not classified based on available information.

Product:

Remarks : This information is not available.

Components:**N-methyl pyrrolidone:**

Carcinogenicity - Assessment : Did not show carcinogenic effects in animal experiments.

Reproductive toxicity

May damage fertility or the unborn child.

Product:

Effects on fertility : Remarks: No data available

STOT - single exposure

May cause respiratory irritation.

Product:

Remarks : No data available

Components:**N-methyl pyrrolidone:**

Assessment : May cause respiratory irritation.

STOT - repeated exposure

Not classified based on available information.

Product:

Remarks : No data available

Repeated dose toxicity**Components:****N-methyl pyrrolidone:**

Species	:	Rat
NOAEL	:	3000 ppm
Application Route	:	Oral
Exposure time	:	90 d

Aspiration toxicity

Not classified based on available information.

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Product:

No aspiration toxicity classification

Experience with human exposure**Product:**

Inhalation	:	Remarks: May cause respiratory irritation.
Skin contact	:	Remarks: Causes skin irritation.
Eye contact	:	Remarks: Causes serious eye irritation.
Ingestion	:	Remarks: None known.

Further information**Product:**

Remarks : None known.

Section 12: Ecological information**Toxicity****Components:****N-methyl pyrrolidone:**

Toxicity to fish	:	LC50 (Oncorhynchus mykiss (rainbow trout)): > 500 mg/l Exposure time: 96 h
Toxicity to daphnia and other aquatic invertebrates	:	LC50 (Daphnia magna (Water flea)): 955 - 1,259 mg/l Exposure time: 96 h NOEC: (Daphnia magna (Water flea)): 1,000 mg/l Exposure time: 24 h

Persistence and degradability**Components:****N-methyl pyrrolidone:**

Biodegradability : Result: Readily biodegradable.

Bioaccumulative potential**Components:****N-methyl pyrrolidone:**

Bioaccumulation : Remarks: Bioaccumulation is unlikely.

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Mobility in soil**Product:**

Distribution among environmental compartments : Koc: 20.94

Other adverse effects

No data available

Section 13: Disposal considerations**Disposal methods**

Waste from residues : Dispose of in accordance with local regulations.

Section 14: Transport information**International Regulations****IATA-DGR**

UN/ID No.	: Not applicable
UN proper shipping name	: Not applicable
Transport hazard class(es)	: Not applicable
Subsidiary risk	: Not applicable
Packing group	: Not applicable
Labels	: Not applicable
Packing instruction (cargo aircraft)	: Not applicable
Packing instruction (passenger aircraft)	: Not applicable

IMDG-Code

UN number	: Not applicable
UN proper shipping name	: Not applicable
Transport hazard class(es)	: Not applicable
Subsidiary risk	: Not applicable
Packing group	: Not applicable
Labels	: Not applicable
EmS Code	: Not applicable
Marine pollutant	: Not applicable

Transport in bulk according to IMO instruments

Not applicable for product as supplied.

Special precautions for user

Remarks : not regulated

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Section 15: Regulatory information**Safety, health and environmental regulations specific for the product in question**

Workplace Safety and Health Act and Workplace Safety and Health (General Provisions) Regulations: This product is subjected to the SDS, labelling, PEL and other requirements in the Act/Regulations.

Environmental Protection and Management Act and : Not applicable
Environmental Protection and Management (Hazardous Substances) Regulations

The components of this product are reported in the following inventories:

TCSI	: On the inventory, or in compliance with the inventory
TSCA	: All substances listed as active on the TSCA inventory
AIRC	: On the inventory, or in compliance with the inventory
DSL	: All components of this product are on the Canadian DSL
ENCS	: On the inventory, or in compliance with the inventory
ISHL	: On the inventory, or in compliance with the inventory
KECI	: On the inventory, or in compliance with the inventory
PICCS	: On the inventory, or in compliance with the inventory
IECSC	: On the inventory, or in compliance with the inventory
NZIoC	: On the inventory, or in compliance with the inventory
TECI	: On the inventory, or in compliance with the inventory

Section 16: Other information

Revision Date	: 23.01.2025
Date format	: dd.mm.yyyy

Full text of other abbreviations

AIRC - Australian Inventory of Industrial Chemicals; ANTT - National Agency for Transport by Land of Brazil; ASTM - American Society for the Testing of Materials; bw - Body weight; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with

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x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; ERG - Emergency Response Guide; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; Nch - Chilean Norm; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NOM - Official Mexican Norm; NTP - National Toxicology Program; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TDG - Transportation of Dangerous Goods; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative; WHMIS - Workplace Hazardous Materials Information System

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

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